

Christopher A. Halsch

Postdoctoral Associate

Department of Biological Sciences • Binghamton University

chalsch@binghamton.edu • (415) 246-9157 • <https://chrishalsch.com>

Appointments

Binghamton University
Postdoctoral Associate

2024 – Present

Education

University of Nevada, Reno

2018 – 23

Doctorate - *Ecology, Evolution, and Conservation Biology*

Advisor: Matthew L. Forister

Committee: Sarah Bisbing, Lee Dyer, James Fordyce, Perry Williams

University of California, Irvine

2011– 15

Bachelor of Science – *Earth System Science*

Bachelor of Science – *Ecology and Evolution*

Grants, Scholarships, and Awards

College of Science Outstanding Graduate Assistant Award (\$500)

2022

- University of Nevada, Reno

USDA AFRI NIFA Predoctoral Fellowship (\$117,700)

2022

- U.S. Department of Agriculture

Centennial Pollinator Fellowship (\$2,500)

2021

- The Garden Club of America

Hitchcock Fellowship (\$13,300)

2021

- Hitchcock Center for Chemical Ecology

Jerry and Betty Wilson Scholarship (\$4,000)

2021

- University of Nevada, Reno

Joan Mosenthal DeWind Award (\$5,000)

2021

- Xerces Society for Invertebrate Conservation

Ron Leuschner Memorial Fund for Research on the Lepidoptera (\$500)

2021

- Lepidopterists Society of America

Ben & Beatrice Edwards Biology Scholarship (\$1,200)

2020 – 21

- University of Nevada, Reno

Outstanding Graduate Student Scholarship (\$500)

2020

- University of Nevada, Reno

Graduate Research Fellowship Program (Honorable Mention)

2020

- National Science Foundation

Publications (in reverse chronological order)

*Denotes undergraduate student

28. **Halsch, C.A.**, Connors, C.J., Grames, E.M., Forister, M.L., Stireman III, J.O., Wagner, D.L., Dyer, L.A. Knowledge Gained and Biodiversity Lost in the World's Best Studied Tropical Forest. *In review*.
27. Grames, E.M.,... **Halsch, C.A.**,... Wagner, D.L. The State of the World's Insects: Summary for Policymakers. *In review*.
26. Wagner, D.L.,... **Halsch, C.A.**,... Grames, E.M. The Global State of Insect Biodiveristy. *In review*.
25. **Halsch, C.A.**,... Wagner, D.L. Drivers of Insect Decline: Direct, Indirect, and Combined Pressures. *In review*.
24. Ware, J.L.,... **Halsch, C.A.**,... Grames, E.M. Insects and Quality of Human Life. *In review*
23. Grames, E.M.,... **Halsch, C.A.**,... Ware, J.L. Insects' Contributions to People: The role insects play in supporting and regulating ecosystem functions and provisioning ecosystem services. *In review*.
22. Bahlai, C.A.,... **Halsch, C.A.**,... Grames, E.M. Future Pathways: Projecting insect responses to global change. *In review*.
21. Elphick, C.S.,... **Halsch, C.A.**,... Grames, E.M. Options for Action and Policy to Conserve Insects. *In review*.
20. **Halsch, C.A.**, Forister, M.L., Grames, E.M. 2026. Dispersal Mediates Metapopulation Response to Local and Regional Stressors. *Insect Conservation and Diversity*. In press.
19. Cushman, J.H., Macdonald, J.A., **Halsch, C.A.**, Refsland, T.K., Steed, B.E., Dilts, T.E. 2026. Decadal Drivers of Quaking Aspen Decline in the Western United States. *Journal of Ecology* 114(6):e70367. doi: 10.1111/1365-2745.70367
18. **Halsch, C.A.**, Nice, C.C., Bell, K.L., Fordyce, J.A., Forister, M.L., Garcia-Costoya, G., Shapiro, A.M., Grames, E.M. 2026. Allochronic Isolation Between Sympatric Populations of an Alpine Butterfly. *Evolution* 80(5):1102-1110. doi: 10.1093/evolut/qpag039
17. Foster, E.M.*, Dombroskie, J.J., **Halsch, C.A.**, Powell, T.H.Q., Grames, E.M. 2026. Phenological Shifts and Increases in Voltinism Within a Moth Community Over a Century of Anthropogenic Change. *Ecology* 107:e70328. doi: 10.1002/ecy.70328
16. Reis, G.A., Forister, M.L., **Halsch, C.A.**, Dittmore, C.M., Shapiro, A.M., Gompert, Z. 2026. Temporal Occupancy Distributions Reveal Diverse Responses to Climatic Variation in Montane Butterflies. *Ecology* 107:e70297. doi: 10.1002/ecy.70297
15. Christensen, T., **Halsch, C.A.**, Dyer, L., Smilanich, A.M., Shapiro, A.M., Forister, M.L. 2025. Specialized Flower Visitation in Montane Butterflies is Associated with Positive Population Trajectories Over Time. *Ecology* 106:e70236. doi: 10.1002/ecy.70236

14. Dittmore, C.M., Anderson, A., Code, A., Lenard, A., Douglas, M.R., **Halsch, C.A.**, Forister, M.L. 2025. Pesticide Contamination of Two Urban Areas has Implications for Insect Conservation and Green Space Management. *Environmental Toxicology and Chemistry*. doi: 10.1093/etjnl/vgaf218
13. **Halsch, C.A.**, Shapiro, A.M., Forister, M.L., Grames, E.M. 2025. Shifting Baselines in North America's Longest-Running Butterfly Monitoring Program. *Conservation Letters* 18:e13116. doi: 10.1111/conl.13116
12. **Halsch, C.A.**, Elphick, C.S., Bahlai, C.A., Forister, M.L., Wagner, D.L., Ware, D.L., Grames, E.M. 2025. Meta-synthesis Reveals Interconnections Among Drivers of Insect Biodiversity Loss. *Bioscience* 75:448-456. doi: 10.1093/biosci/biaf034
11. **Halsch, C.A.**, Shapiro, A.M., Thorne, J.H., Rodman, K.C., Parra, A., Dyer, L.A., Gompert, Z., Smilanich A.M., and Forister, M.L. 2024. Thirty-six Years of Butterflies, Snow, and Plant Productivity Reveal Negative Impacts of Warmer Winters and Increased Productivity on Montane Species. *Global Change Biology* 30:e17044. doi: 10.1111/gcb.17044.
10. Forister, M.L., Grames, E.M., **Halsch, C.A.**, Burls, K.J., Carroll, C.F., Bell, K.L., Jahner, J.P., Bradford, T., Zhang, J., Cong, Q., Grishin, N.V., Glassberg, J., Shapiro, A.M., and Riecke, T.V. 2023. Assessing Risk for Butterflies in the Context of Climate Change, Demographic Uncertainty, and Heterogenous Data Sources. *Ecological Monographs* 93:e1584. doi: 10.1002/ecm.1584
9. Forister M.L., Black S.H., Elphick C.S., Grames E.M., **Halsch C.A.**, Schultz C.B., Wagner, D.L. 2023. Missing the Bigger Picture: Why Insect Monitoring Programs are Limited in their Ability to Document the Effects of Habitat Loss. *Conservation Letters* 16:e12951. doi: 10.1111/conl.12951
8. **Halsch, C.A.**, Zullo, D.J.*, and Forister, M.L. 2023. Additive and Interactive Pressures of Anthropogenic Stressors on an Insect Herbivore. *Proceedings of the Royal Academy B* 290: 2022243. doi: 10.1098/rspb.2022.2431
7. **Halsch, C.A.**, Hoyle, S.M., Code, A., Fordyce, J.A., Forister, M.L. (2022) Milkweed Plants Bought at Nurseries May Expose Monarch Caterpillars to Harmful Pesticide Residues. *Biological Conservation* 273:109699. doi: 10.1016/j.biocon.2022.109699
6. Forister, M.L., **Halsch, C.A.**, Nice, C.C., Fordyce, J.A., Dilts, T.E., Oliver, J.C., Prudic, K.L., Shapiro, A.M., Wilson, J.K., and Glassberg, J. 2021. Fewer Butterflies Seen by Community Scientists Across the Warming and Drying Landscapes of the American West. *Science* 371:1042-1045. doi: 10.1126/science.abe5585
5. **Halsch, C.A.**, Shapiro, A.M., Fordyce, J.A., Nice, C.C., Thorne, J.H., Waetjen, D.P., and Forister, M.L. 2021. Insects and Recent Climate Change. *Proceedings of the National Academy of Sciences* 118:e2002543117. doi: 10.1073/pnas.2002543117
4. **Halsch, C.A.**, Code, A., Hoyle, S.M., Fordyce, J.A., Baert, N., and Forister, M.L. 2020. Pesticide Contamination of Milkweeds Across the Agricultural, Urban, and Open Spaces of Low-Elevation Northern California. *Frontiers in Ecology and Evolution* 8. doi: 10.3389/fevo.2020.00162

3. **Halsch, C.A.**, Shapiro, A.M., Thorne, J.H., Waetjen, D.P., and Forister, M.L. 2020. A Winner in the Anthropocene: Changing Host Plant Distribution Explains Geographic Range Expansion in the Gulf Fritillary Butterfly. *Ecological Entomology* 45:652-662. doi: 10.1111/een.12845
2. Kimball, S., Long, J., Ludovise, S., Ta, P., Schmidt, K., **Halsch, C.A.**, and Magliano, K. 2019. Impacts of Competition and Herbivory on Native Plants in a Community-Engaged, Adaptively Managed Restoration Experiment. *Conservation Science and Practice* 1:e122. doi: 10.1111/csp2.122
1. Tamura, N.*, Lulow, M.E., **Halsch, C.A.***, Major, M.R., Balazs, K.R., Austin, P., Huxman, T.E., and Kimball, S. 2017. Effectiveness of Seed Sowing Techniques for Sloped Restoration Sites. *Restoration Ecology* 25:942-952. doi: 10.1111/rec.12515

Teaching Experience

- BIOL 750 – Research Design** Spring 2020 - 22
 Developed lessons and led labs introducing graduate students to R coding and statistical analysis for ecological data.
- EECB 751 – Philosophy of Science** Fall 2020
 Designed and implemented a graduate course on the philosophy of science, focusing on topics related to ecology, evolution, and conservation biology.
- BIOL 437 – Entomology** Spring 2019
 Organized and led labs on insect identification and taxonomy.
- Crystal Cove Conservancy, Newport Beach, CA** Spring 2016-18
 Designed and implemented participatory science education programs for K-12 students in partnership with California State Parks and University of California, Irvine researchers.

Professional Presentations

Seminars and Plenaries

- Halsch, C.A.** 2026. *Western montane butterflies in a changing world*. University of Washington, Department of Biology seminar. Seattle, Washington, USA.
- Halsch, C.A.** 2026. *From monitoring to genetics: different perspectives on insect biodiversity loss*. Wright State University, Department of Biological Sciences seminar. Dayton, Ohio, USA.
- Halsch, C.A.** 2026. *Loss of snow and declines of butterflies in the Sierra Nevada*. Status of Insects Climate Change Working Group. Virtual.
- Halsch, C.A.** 2026. *Insect Biodiversity in a Changing World*. Syracuse University, Department of Biology seminar. Syracuse, New York, USA.
- Halsch, C.A.** 2025. *Population trends and adaptation in a changing world: insights from western butterflies*. Binghamton University, Department of Biological Sciences seminar. Binghamton, New York, USA.

Halsch, C.A. 2025. *Direct and indirect effects of climate on montane butterflies*. University of Nevada, Reno EECB program colloquium. Reno, Nevada, USA.

Halsch, C.A., Shapiro, A.M., Parra, A.S., Rodman, K.C., Thorne, J.H., Forister, M.L. 2023. *Direct and indirect effects of climate on montane butterflies*. Butterfly Conservation symposium plenary speaker. Bedford, UK.

Invited Presentations

Halsch, C. A., Grames, E.M. 2025. *Insect overwintering modulates responses to temperature change*. Entomological Society of America. Portland, OR, USA.

Halsch, C.A., Forister, M.L., Grames, E.M. 2025. *One day to rule them all*. Status of Insects Research Coordination Network annual meeting. Virtual meeting.

Halsch, C.A., Grames, E.M. 2024. *Death by 852 cuts: connections among drivers of insect decline*. Entomological Society of America. Phoenix, AZ, USA.

Halsch, C.A., Forister, M.L., Grames, E.M. 2024. *Considering the risk of pesticide exposure across already stressed populations*. International Congress of Entomology. Kyoto, Japan.

Halsch, C.A. 2023. *Pesticide risk to butterflies in the western US*. Entomological Society of America Pacific Branch. Seattle, WA, USA.

Halsch, C.A., Selvaggio, S., Code, A. *Rethinking Nursery Production Practices for Safe Pollinator Plants*. 2023. Tri-State Green Industry Conference. Glendale, OH, USA.

Halsch, C. A., Forister, M.L., Grames, E.M., Burls, K.J., Carroll, C.F., Bell, K.L., Jahner, J.P., Bradford, T., Zhang, J., Cong, Q., Grishin, N.V., Glassberg, J., Shapiro, A.M., and Riecke, T. V. 2022. *Integrating heterogenous data sources to assess the status and risk of butterflies to Anthropogenic threats in the western United States*. International Congress of Entomology. Helsinki, Finland.

Halsch, C.A., Shapiro, A.M., Parra, A.S., Rodman, K.C., Thorne, J.H., Forister, M.L. 2022. *Separating the direct and indirect effects of climate change on butterflies in the Sierra Nevada, CA using remote sensing data*. Entomological Society of America Pacific Branch. Santa Rosa, CA, USA.

Halsch, C.A., Shapiro, A.M., Forister, M.L. 2020. *Understanding global change and butterflies with Western North America's longest-running monitoring study*. Entomological Society of America. Virtual meeting.

Contributed Presentations

Halsch, C.A. 2023. *The direct and indirect effects of climate stressors on montane butterflies*. Ecological Society of America. Portland, OR, USA.

Halsch, C.A., Shapiro, A.M., Parra, A.S., Thorne, J.H., Forister, M.L. 2022. *Climate change and butterflies: Can we use long-term data to separate direct effects on individuals from plant-mediated indirect effects?* Entomological Society of America. Denver, CO, USA.

Halsch, C.A., Shapiro, A.M., Forister, M.L. 2019. *An Expanding Fritness Landscape: Minimum Temperatures, Host Plant Distribution, and the Expansion of the Gulf Fritillary*. Entomological Society of America. St. Louis, MO, USA.

Halsch, C.A., Shapiro, A.M., Forister, M.L. 2019. *The Spatial and Temporal Story of the Expanding Gulf Fritillary Butterfly*. The Lepidopterist's Society. Davis, CA, USA.

Professional Service

Organized Symposia

Grames, E.M., Millard, J., **Halsch, C.A.,** Elphick, C. 2025. *Understanding Drivers of Insect Biodiversity Loss Through Synthesis*. Entomological Society of America.

Halsch, C.A., Grames, E.M. 2024. *Insect Decline Research Coordination Network Symposium and Workshop*. Entomological Society of America.

Halsch, C.A., Grames, E.M., Wagner, D.L. 2022. *Insect decline in the Anthropocene*. International Congress of Entomology.

Journal Reviews

Annals of the Entomological Society of America, Biodiversity and Conservation, Biological Conservation, Biological Reviews, Communications Biology, Current Opinion in Insect Science, Diversity and Distributions, Ecological Applications, Ecological Entomology, Ecology, Ecology Letters, Ecosphere, Global Change Biology, Global Ecology and Biogeography, Insect Conservation and Diversity, Journal of Biogeography, Journal of Insect Conservation, Journal of Pest Science, Landscape Ecology, PeerJ

Other Professional Service

Handling editor – Journal of Lost Species	2025-Present
Grant reviewer – NSF	2025
Board member – Nevada Bugs and Butterflies	2020-22
Student President – UNR EECB Graduate Program	2020-21
President – Plant-Animal Interactions Club	2018-20

Public Outreach

Community Outreach

Leader – North America Butterfly Association butterfly counts 2019-Present
Butterfly monitoring programs for public engagement with butterfly research.

Co-founder – Nerd Nite, Reno, Community Lecture Series 2019-23
Monthly lecture series for large audiences, featuring speakers presenting research.

Public Presentations

Halsch, C.A. 2026. *Where the wild things are*. Nerd Nite Reno public lecture series.

Halsch, C.A., Grames, E.M. 2024. *Death by 852 cuts: connections among drivers of insect decline*. Status of Insects Research Coordination Network webinar series.

Halsch, C.A. 2023. *Beyond Basic Butterflies*. Nerd Nite Reno public lecture series.

Halsch, C.A. 2022. *The state of butterflies in the western US*. University of Nevada Extension: Cultivating healthy plants: An IPM webinar series.

Halsch, C.A. 2022. *Looking for pesticides in milkweeds sold in nurseries*. Xerces Society for Invertebrate Conservation.

Halsch, C.A. 2021. *Contamination of Marginal Spaces and the Role of Pesticides in Butterfly Declines*. Western Hummingbird Partnership.